TRR356 WORKSHOP · PART 1

Bacterial phylogenomics

Part 1: Pangenomes, genome similarity and gene content

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Who is in the room?

Haim Ashkenazy

Postdoc, Department of Molecular Biology (Weigel), MPI for Biology Tübingen

- Microbe–microbe–host interactions in *Arabidopsis*
- Molecular evolution and comparative genomics tools: ConSurf, GUIDANCE, PanGene-O-Meter, GLOOME and others
- PhD in molecular evolution and bioinformatics, Tel Aviv University, Israel

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Google Scholar: scholar.google.com/citations?user=fQ3SLLgAAAAJ
Bluesky: @haimashkenazy.bsky.social

Your turn (about one minute each)

- Name, group and project
- Which bacteria or system do you work on?
- Do you have genomes, or plan to?
- One thing you want to get out of today



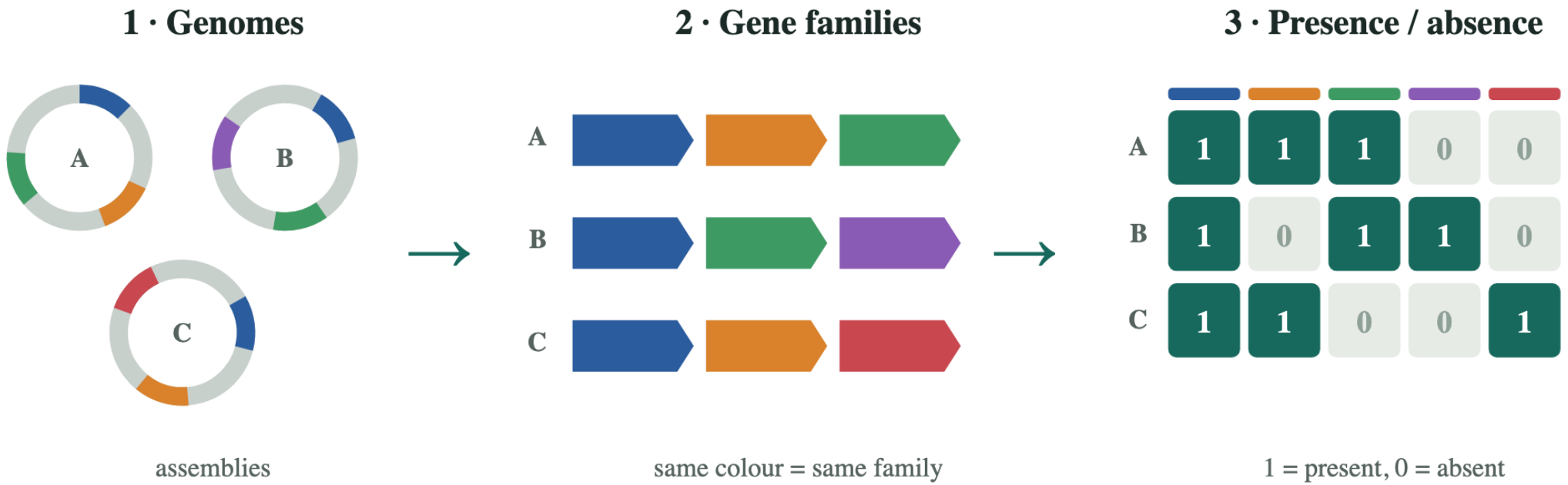
Please bring your own genomes to the afternoon session if you have them. If not, we work on the 63-genome *P. syringae* example.

Agenda

Time	Block	Content
09:30	Part 1 (65 min)	Pangenome concept · ANI, skani, PopPUNK · gene-content similarity (PanGene-O-Meter) · Microbializer
10:35	Break (15 min)	
10:50	Part 2 (70 min)	Gain and loss (GLOOME) · co-evolution (CoPAP) · gene-content GWAS (treeWAS, pyseer)
12:00	Break	until 14:30
14:30	Hands-on and consultation	Our Pseudomonas set: Microbializer results, gene-content similarity, GLOOME and CoPAP, optional GWAS

Materials, handout and data: github.com/HaimAshk/bacterial-phylogenomics-workshop

From genomes to a presence/absence matrix



Pangenome = all gene families seen in at least one genome: here **5**.

Core genome = families present in every genome: here **1** (blue).

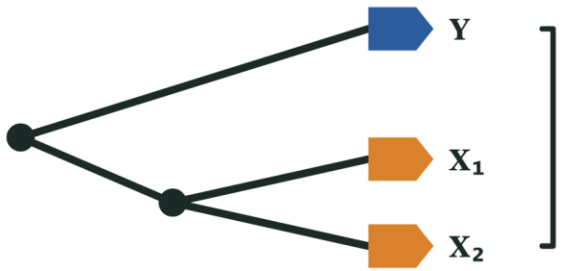
Schematic of the PanGene-O-Meter workflow (Ashkenazy and Weigel 2025), steps 1–3

Homologs, orthologs and paralogs

One gene family, drawn as a gene tree. Two genomes, Y and X; the gene was duplicated in X only (X_1, X_2). ◆ **duplication** · ● **speciation**. Same tree in each panel: the highlighted part is the relationship.

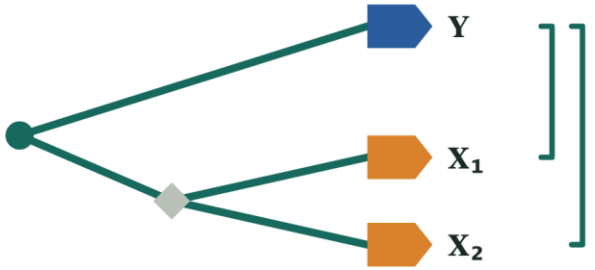
Homologs

All three genes: they share one ancestral gene.



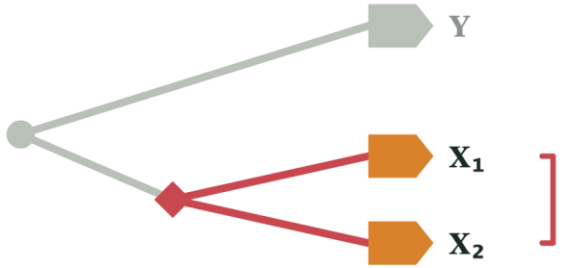
Orthologs: split by speciation

Y–X_1 and **Y–X_2**: the same gene in two genomes. X has two copies; both are orthologs of Y.



Paralogs: split by duplication

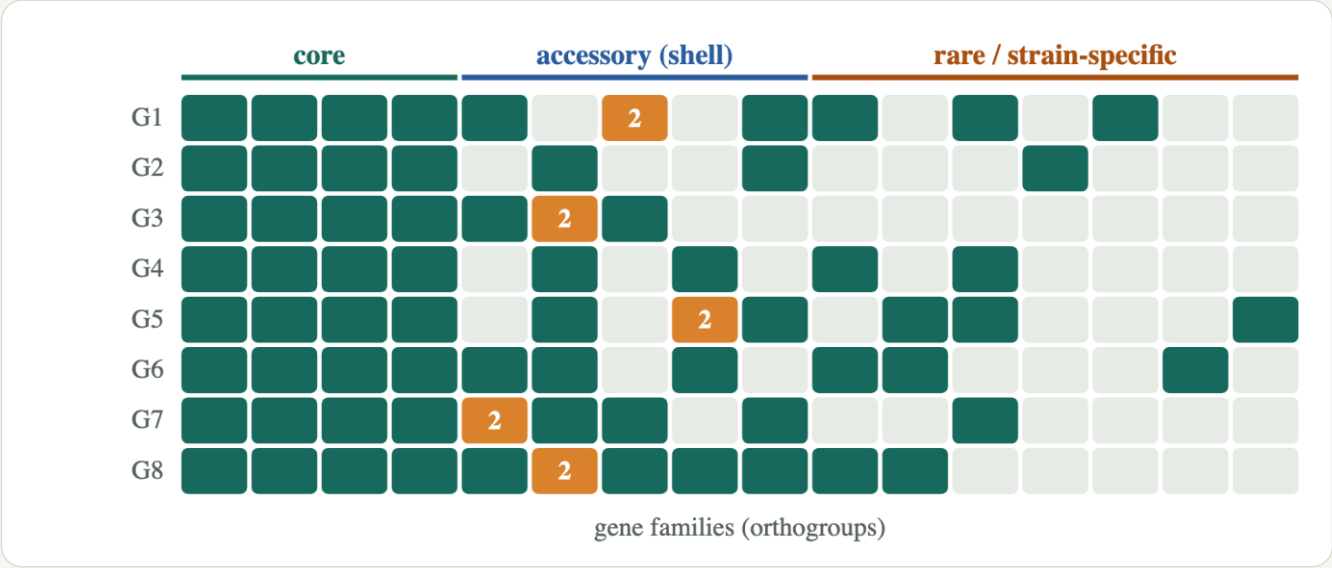
X_1–X_2: two copies of the gene in one genome.



Gene family = orthogroup. One column of the matrix. A recent duplicate stays in the family: that is a **multi-copy** ortholog. The matrix is **inferred**: thresholds, fragmented assemblies and annotation all change it.

Definitions: Fitch (1970); Koonin (2005) Annu Rev Genet 39:309; orthogroups as in Microbializer documentation and OrthoFinder (Emms and Kelly 2019)

The pangenome: core, accessory and rare genes



Gene family

Orthologs, one per genome. **Orange "2"**: a multi-copy family (recent paralogs), still counted as present.

Pangenome

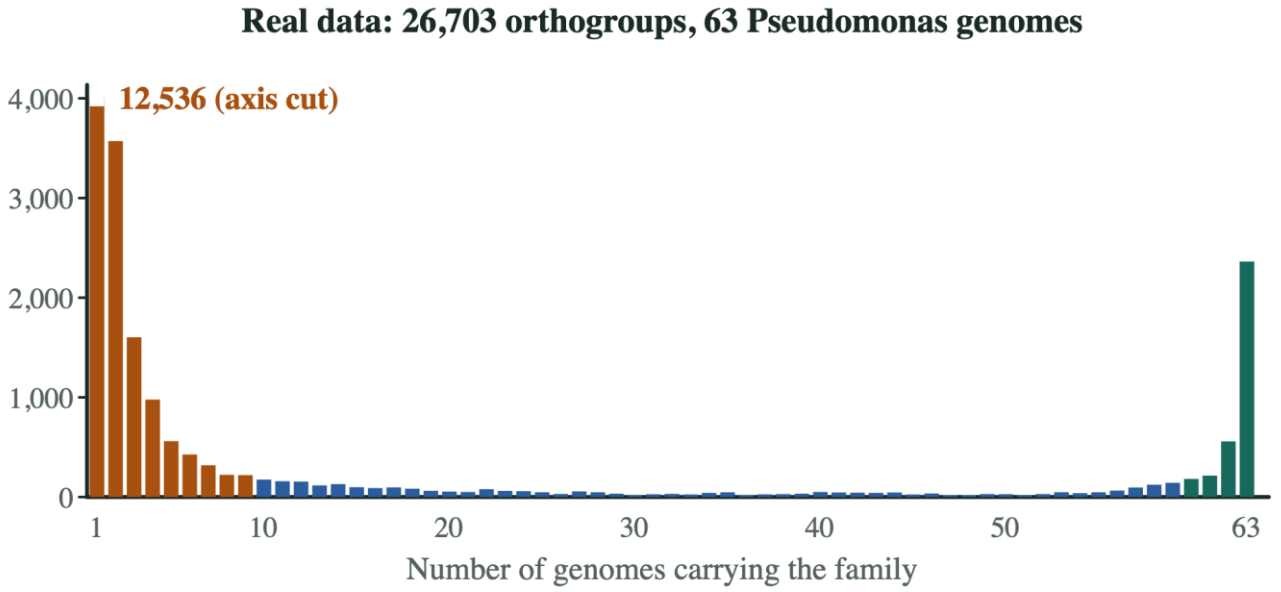
All gene families found in at least one genome of the set.

Core genome

Families in all, or nearly all (e.g. ≥95%), genomes.

Schematic example. Concept: Tettelin et al. (2005) PNAS 102:13950

The U-shaped frequency distribution



Rare: 1–9 genomes

20,431 families (77%), 12,536 in a single genome.

Shell: 10–59 genomes

2,957 families (11%).

Core: ≥95% (60–63)

3,315 families (12%); 2,362 in all 63.

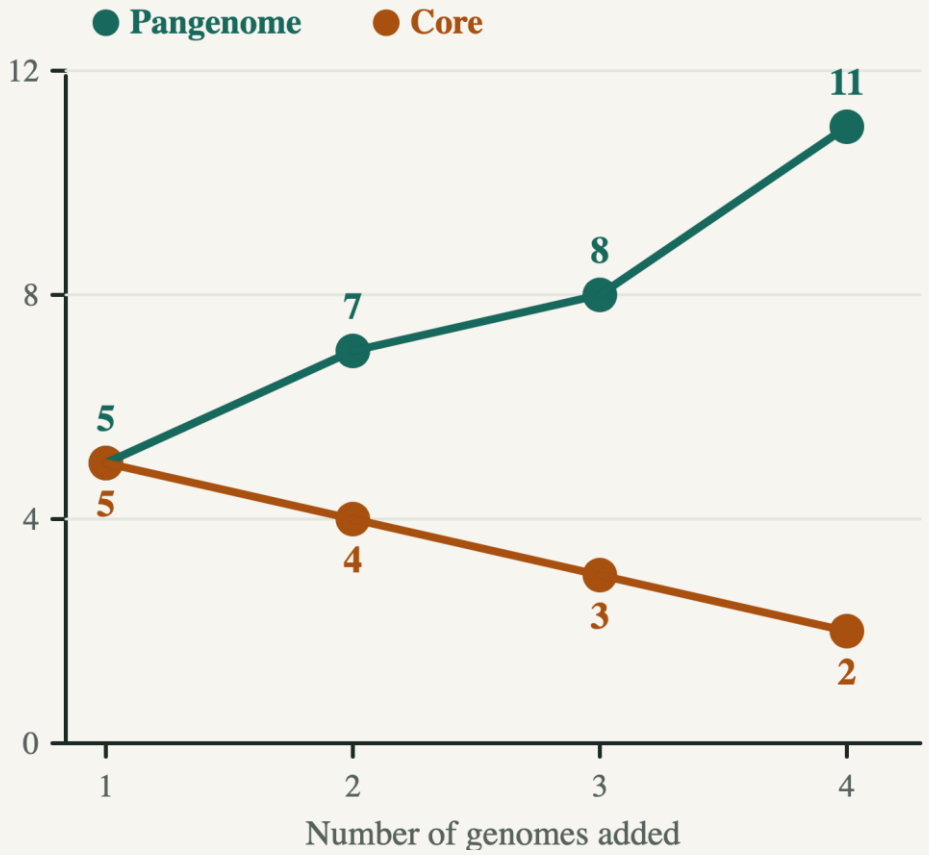
The **U-shape**: many rare families, a core, and few in between. A long left arm is typical of an **open** pangenome. This is the same matrix as before, summed over genomes.

Orthogroup sizes from the author's Microbializer run (orthogroups_annotated.csv); concept: Koonin and Wolf (2008) NAR 36:6688

How a saturation curve is built

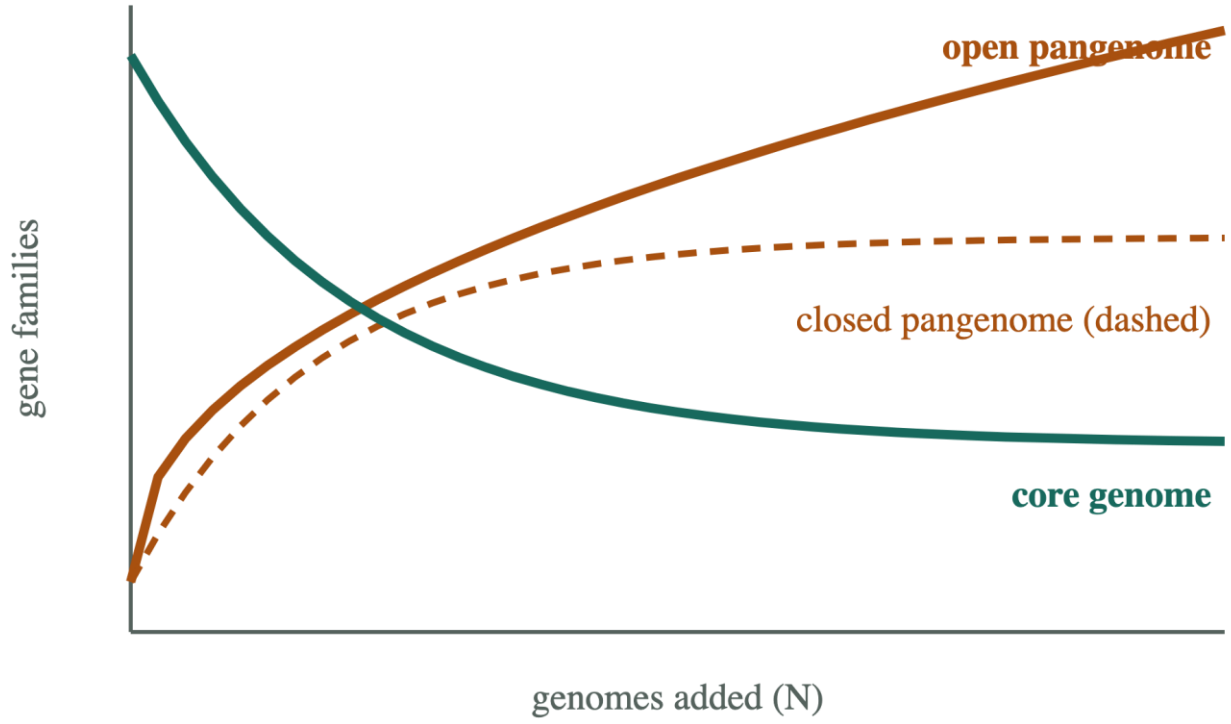
Genome	Gene families	New	Pangenome	Core
G1	a b c d e	5 (–)	5	5
G2	a b c d f g	+2 (f, g)	7	4
G3	a b c h	+1 (h)	8	3
G4	a b i j k	+3 (i, j, k)	11	2

Repeat over **many random orders**, plot the mean. **Slope** of the pangenome curve = new families per added genome; flat = saturated.



Worked example with four toy genomes (letters = gene families; ringed = new to the pangenome). Click to add each genome.

Open and closed pangenomes



How to read it

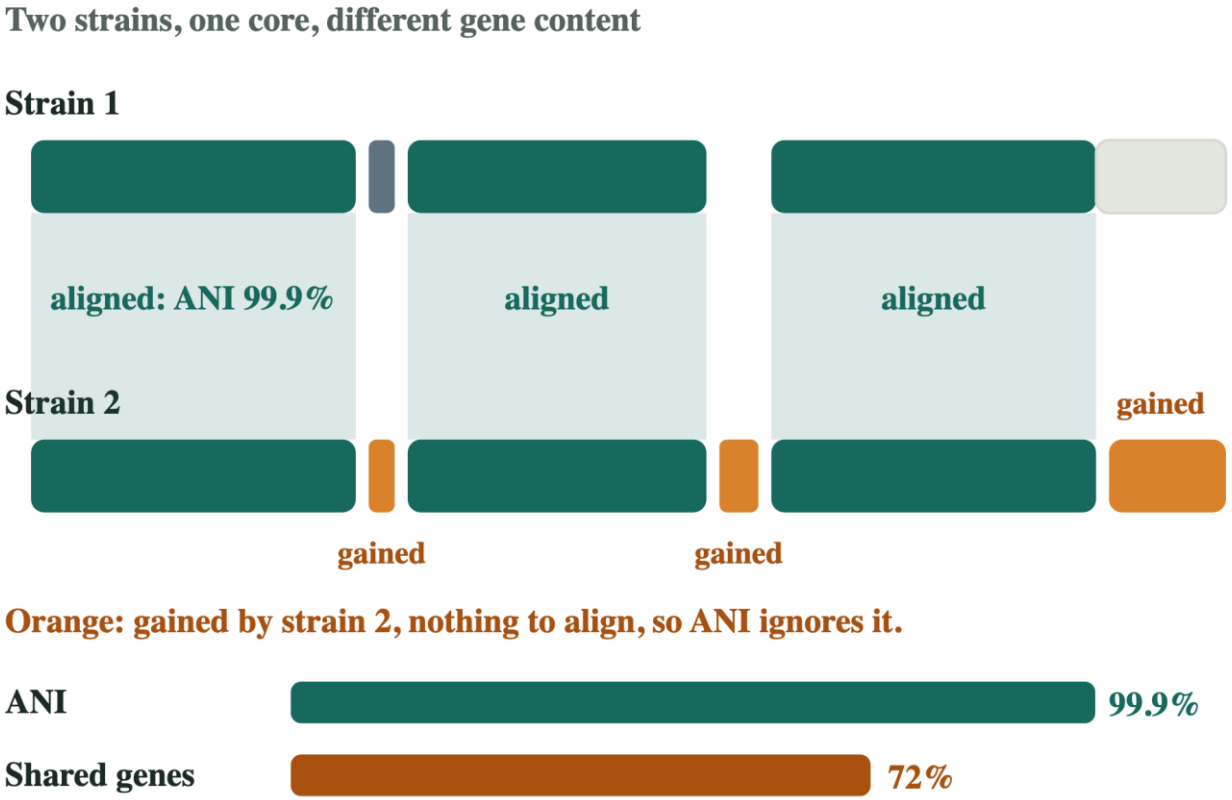
Pangenome curve: distinct families in the first N genomes. **Core curve:** families in all of them. Mean over random genome orders.

Open or closed?

Heaps' law, **n ≈ κ·N^γ**: with γ > 0 every new genome adds families. Wild-leaf *Pseudomonas* and *Sphingomonas*: >95% of families are flexible.

Schematic curves. Tettelin et al. (2008) Curr Opin Microbiol 11:472; leaf numbers from Ashkenazy and Weigel (2025) and the author's talk

What ANI misses: gene-content variation



ANI lens

ANI is computed on the **aligned** part only: **99.9%**. Same organism.

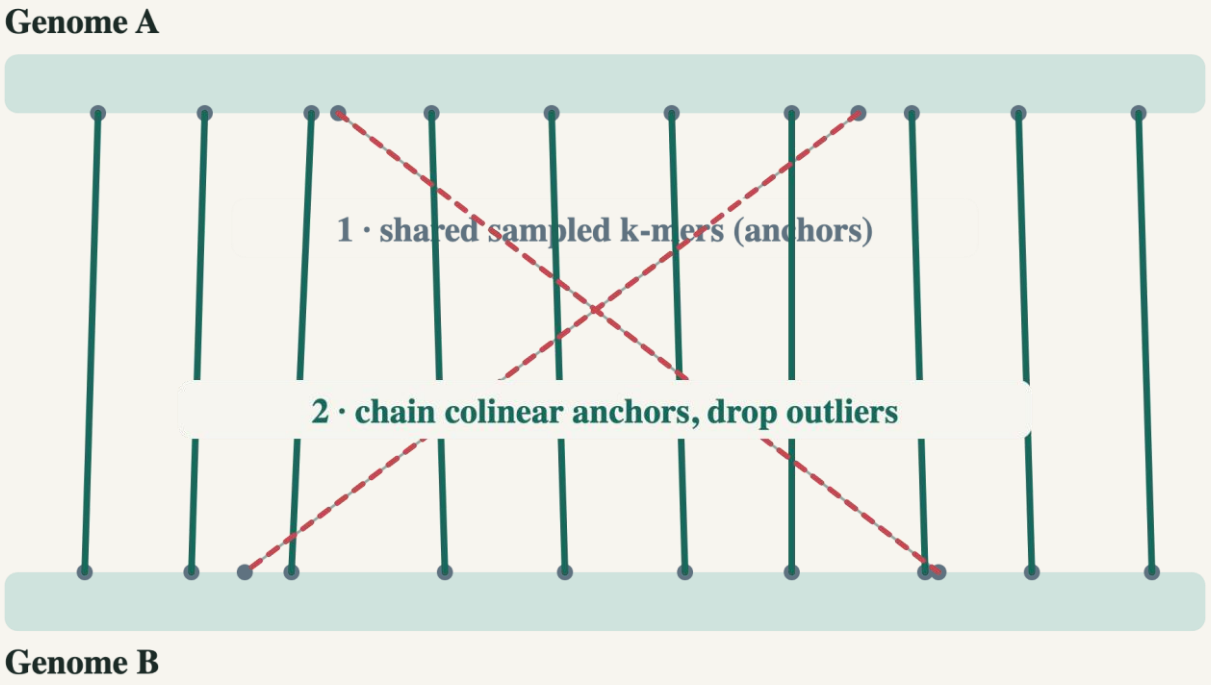
Gene-content lens

The gains and losses outside the aligned part: gene content similarity only **72%**.

From PanGene-O-Meter: “identical” genomes by ANI can still differ in a large share of their genes.

Schematic. Concept and data: Ashkenazy and Weigel (2025), PanGene-O-Meter; example strains from the author’s talk

ANI, and skani: the tool that estimates it



3 · ANI from the chained regions: **ANI 98.7%**, aligned fraction **91% / 89%** (of A / of B).

ANI is the measure

Mean nucleotide identity over the aligned parts of two genomes. Species boundary about **95–96%**.

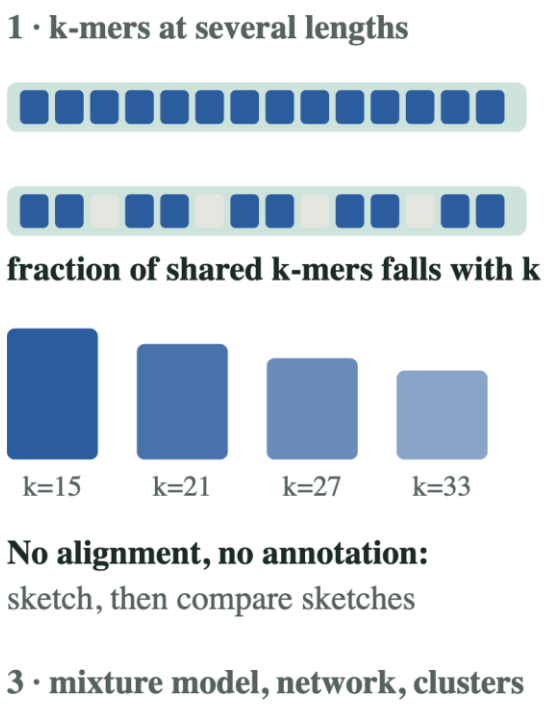
skani is the tool

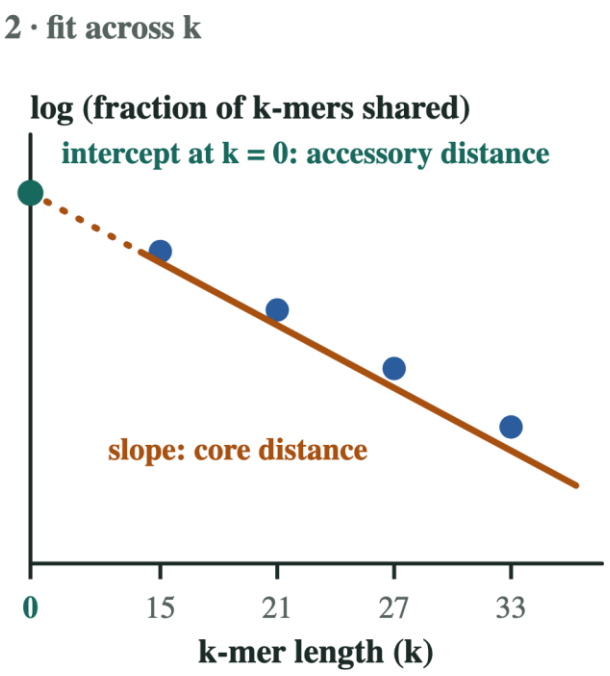
- **Estimates** ANI from sparse k-mer chains: no base-level alignment
- Also reports the **aligned fraction**
- Robust to fragmented, incomplete assemblies
- Fast; reliable above ~82% ANI

Always read **ANI and aligned fraction** together: 99% over 60% of the genome is not the same as 99% over 95%.

Sources: Jain et al. (2018) Nat Commun 9:5114; Shaw and Yu (2023) Nat Methods 20:1661; skani documentation. Schematic; click to step.

PopPUNK: a k-mer tool for core and accessory distance





PopPUNK is a tool

A **k-mer based** method to cluster bacterial genomes into **strains**.

Two distances per pair

- **Core**: from the slope across k (mutations in shared sequence)
- **Accessory**: from the intercept (sequence not shared)

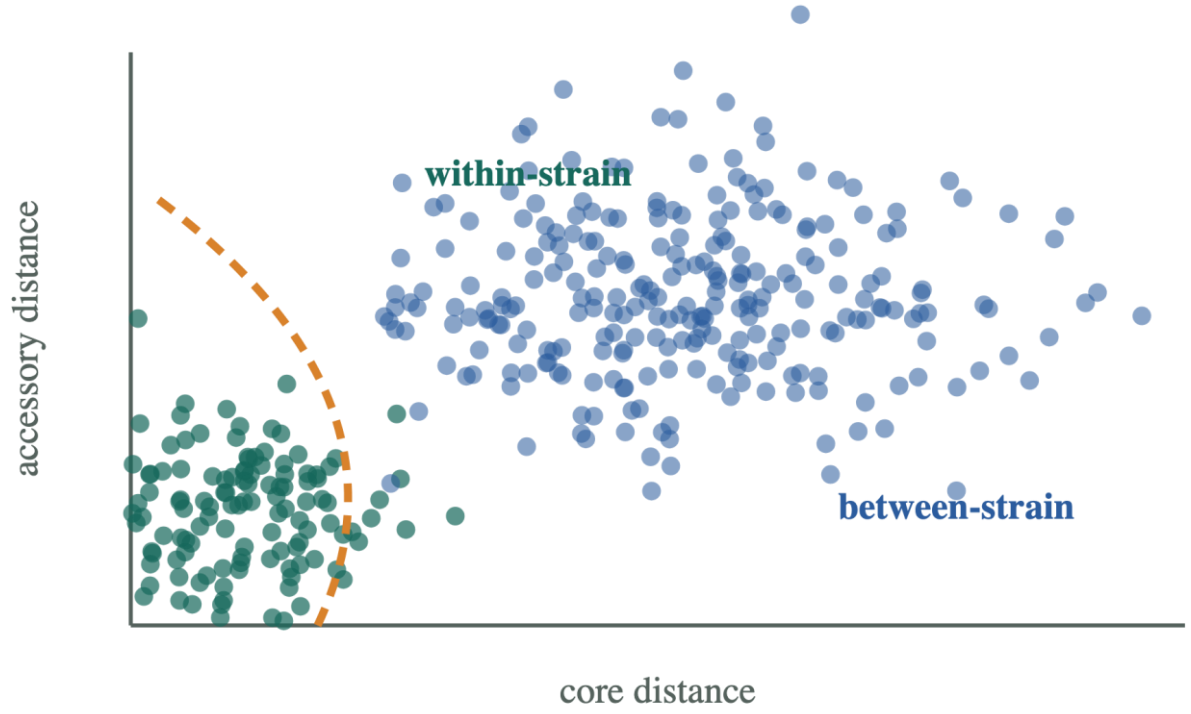
From distances to strains

- Mixture model: within- vs between-strain pairs
- Network components = clusters
- New genomes assigned to an existing database

Source: Lees et al. (2019) Genome Res 29:304; PopPUNK documentation (poppunk.readthedocs.io). Schematic.

SUPPLEMENTARY · skip if short on time

Reading a PopPUNK plot



Reading the plot

- Each point is a genome pair
- Boundary (dashed) sets the strain cut-off
- Accessory distance can be large while core distance is tiny: the **gene-content signal**

PopPUNK clusters are strains. PanGene-O-Meter asks the next question: how much do genes differ *within* a strain?

Schematic, not data. Concept: Lees et al. (2019) Genome Res 29:304

Three ways to compare genomes

	skani (ANI)	PopPUNK	PanGene-O-Meter
Measures	Identity of aligned parts	Core and accessory distance	Shared gene families
Best for	Species identity, dereplication	Strains and lineages	Functional diversity; which genes differ and lineages
Blind spot	Accessory genome	Which genes differ	Quality of gene families

E. coli: gene-content similarity correlates with PopPUNK similarity at r = 0.88, with aligned fraction at r = 0.91, and with ANI only at r = 0.69.

Source: Ashkenazy and Weigel (2025), E. coli, 700 genomes

SUPPLEMENTARY · skip if short on time

K-mers or gene families?

K-mer based

- skani, PopPUNK (and Mash)
- Fast: no gene calling, no annotation
- Sees **all** sequence: genes, **intergenic regions**, regulatory elements, mobile DNA
- But a difference does not say **which gene** or what function

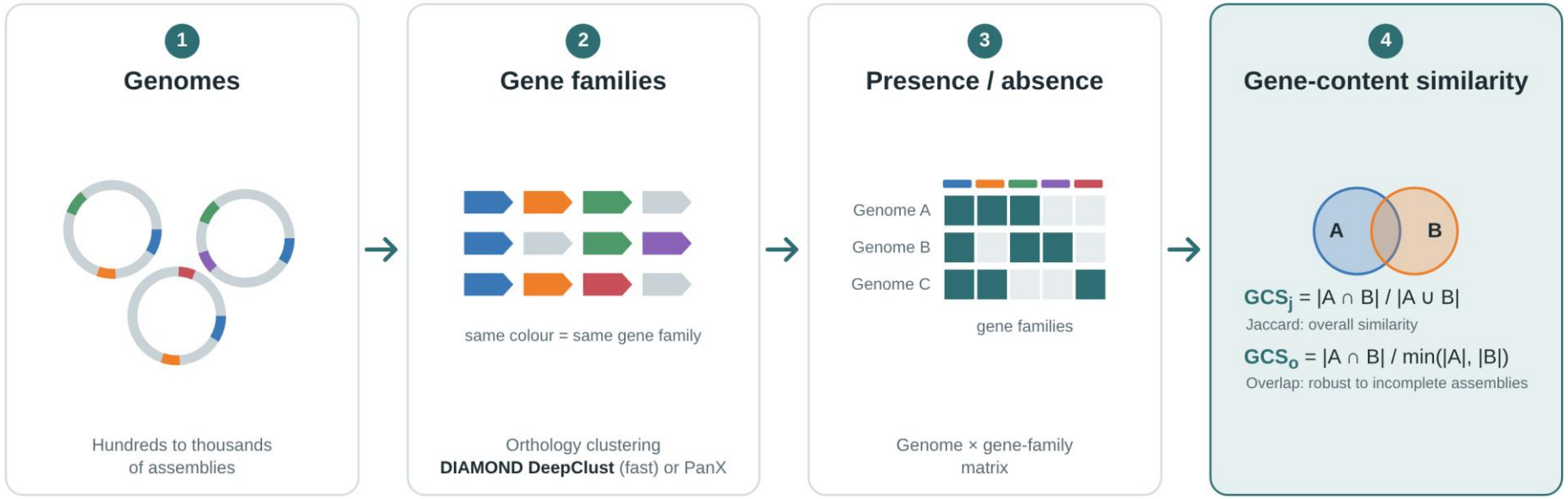
Gene-family based

- PanGene-O-Meter, pangenome tools
- Needs gene prediction and clustering: slower
- Sees **coding genes** only; intergenic variation is invisible
- Every difference points to a **specific family**, with function

K-mers answer “**how different**?”, gene families answer “**what differs**?”. The same trade-off returns in GWAS: k-mer and unitig variants versus gene presence/absence.

Summary by the author; skani: Shaw and Yu (2023); PopPUNK: Lees et al. (2019)

PanGene-O-Meter: genomes by what they carry



WHAT YOU GET

1000s
of genomes

Scales
DIAMOND DeepClust clusters genes fast, scaling easily to thousands of genomes

30%
missing genome

Robust
GCS$_o$ stays accurate even when one genome is 30% incomplete

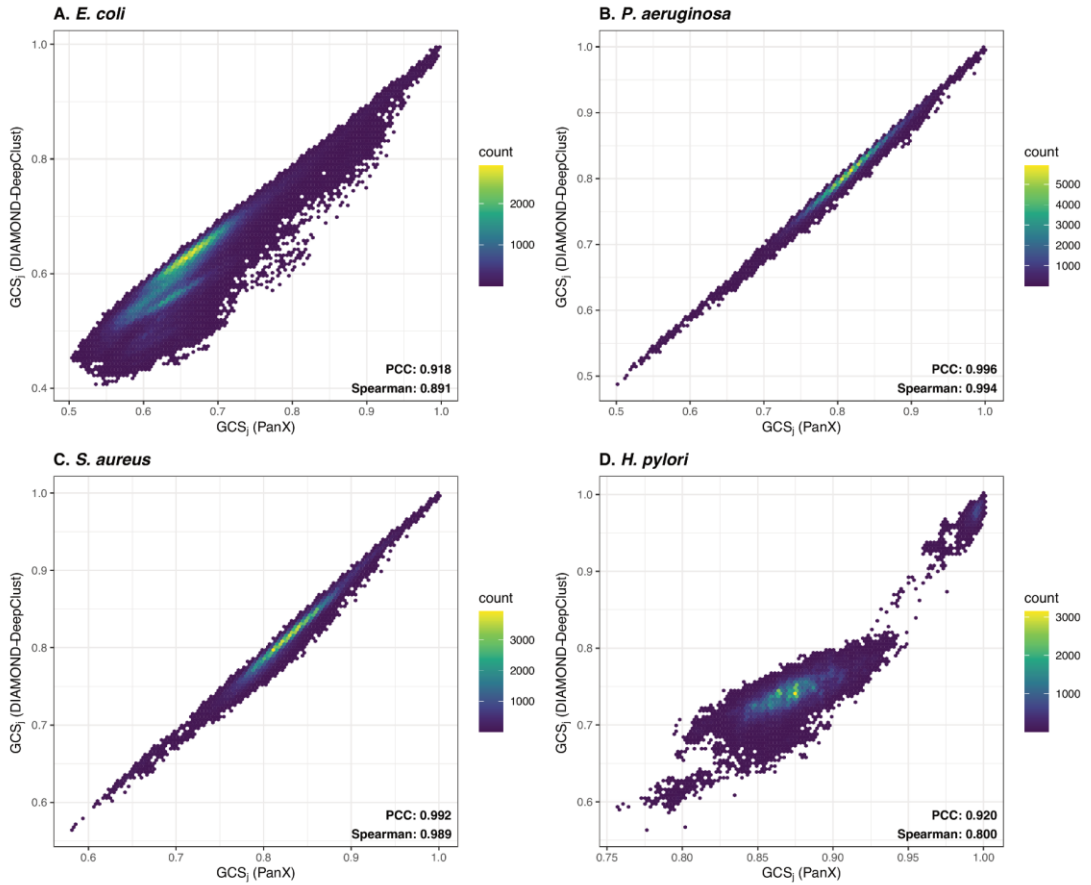
Which genes?
not just how similar

Interpretable
Differences trace back to specific gene families

Source: Ashkenazy and Weigel (2025) PanGene-O-Meter

SUPPLEMENTARY · skip if short on time

Do the gene families matter? Only a little



Fast clustering, same answer

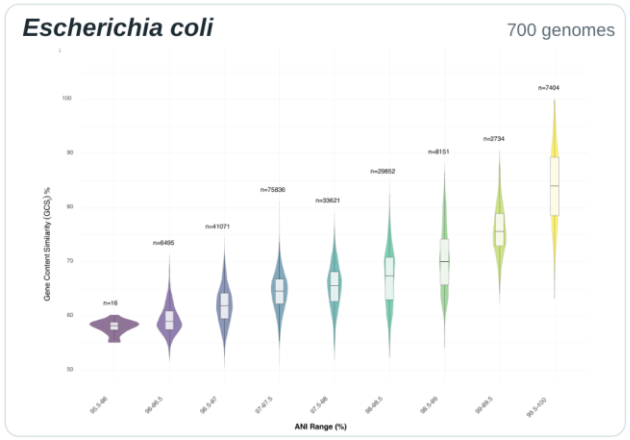
Gene-content similarity from **DIAMOND DeepClust** gene families closely matches the slower **PanX** pangenome.

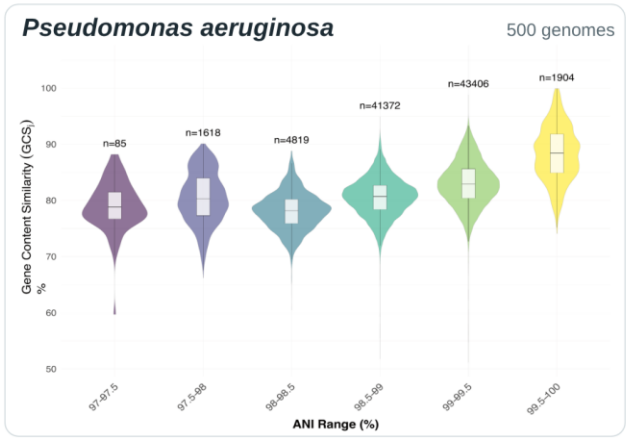
Agreement with PanX (Pearson r)

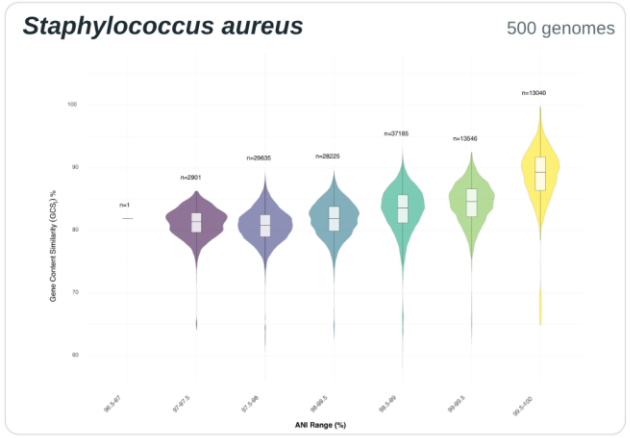
P. aeruginosa	**0.996**
S. aureus	**0.992**
H. pylori	**0.920**
E. coli	**0.918**

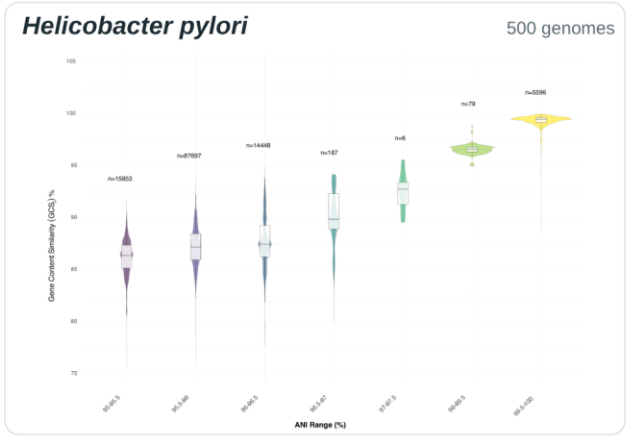
DIAMOND DeepClust (Buchfink et al.) is the default in PanGene-O-Meter, earlier PanX. Ashkenazy and Weigel (2025); Pearson r 0.918–0.996

Near-identical strains differ in gene content









Near-identical ≠ same genes

Even at **99.5–100% ANI**, gene content varies widely in *E. coli*, *P. aeruginosa* and *S. aureus*.

H. pylori is the exception: a stable gene repertoire.

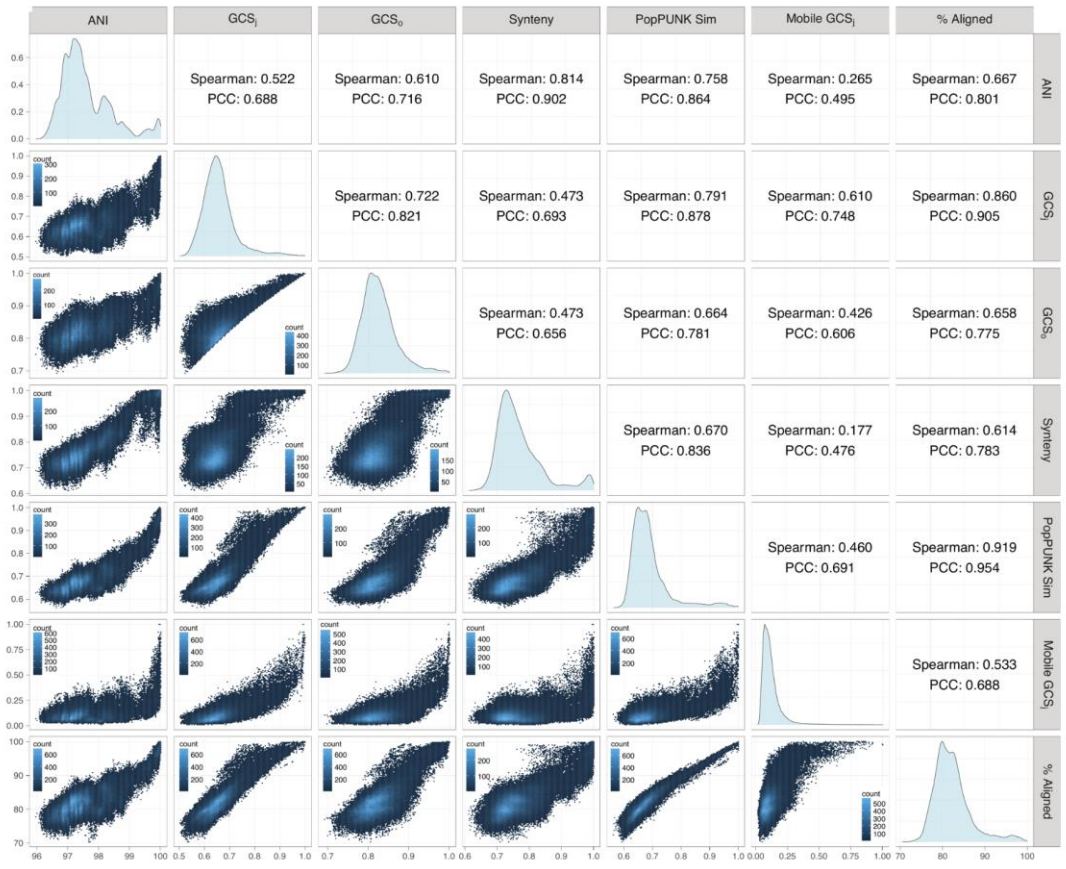
Agrees with established measures

E. coli, GCS_j vs.	Pearson r
PopPUNK similarity	0.88
% genome aligned	0.91
ANI	0.69

Source: Ashkenazy and Weigel (2025); 500–700 genomes per species

SUPPLEMENTARY · skip if short on time

Gene content is not redundant with ANI or PopPUNK



E. coli: all-vs-all comparison of measures

Lower triangle: pairwise genome comparisons; upper triangle: Spearman and Pearson correlations.

Measures

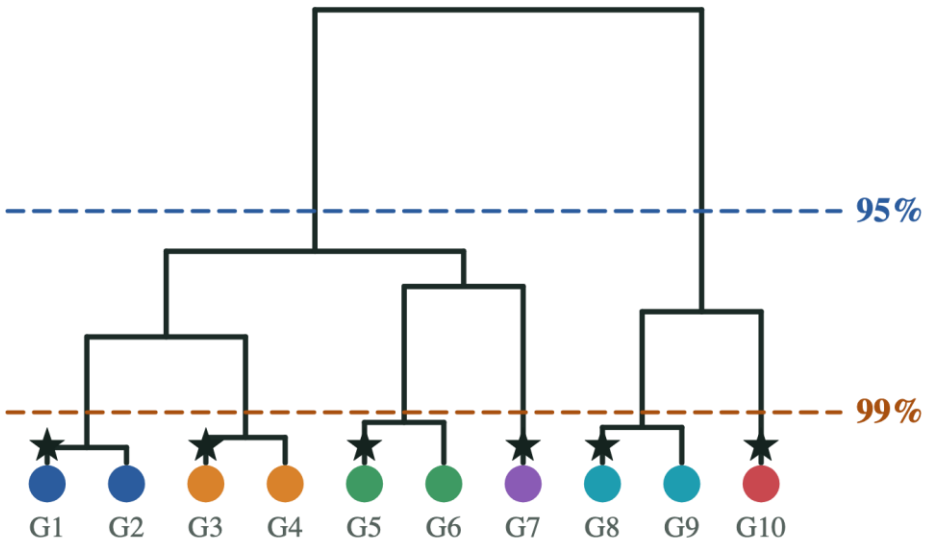
- **ANI**: nucleotide identity of aligned regions
- **GCS$_j$ / GCS$_o$**: gene-content similarity (Jaccard / overlap)
- **Synteny**: gene-order conservation
- **PopPUNK Sim**: k-mer-based similarity
- **Mobile GCS$_j$**: gene content of mobile elements only
- **% Aligned**: fraction of the genomes that aligns

GCS$_j$ tracks PopPUNK (r = 0.88) and % aligned (r = 0.91), and is only moderately correlated with ANI (r = 0.69): it captures what sequence identity misses.

Same analysis for *P. aeruginosa*, *S. aureus* and *H. pylori* in Figs S2–S4.

Source: Ashkenazy and Weigel (2025), E. coli

Grouping genomes by similarity

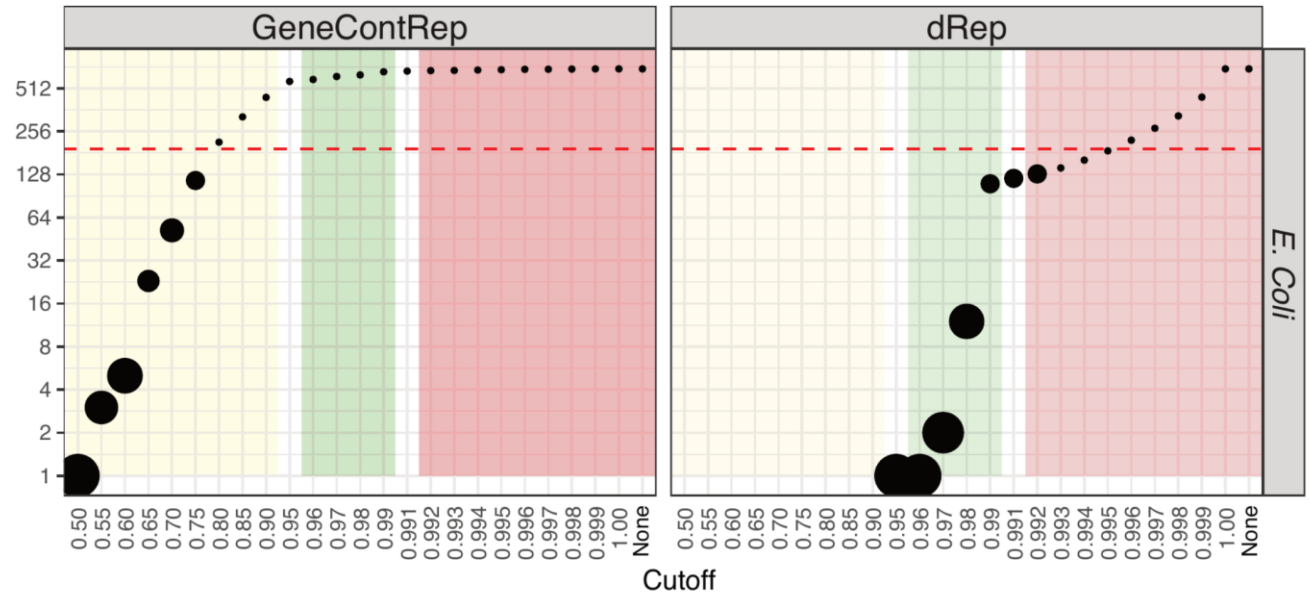


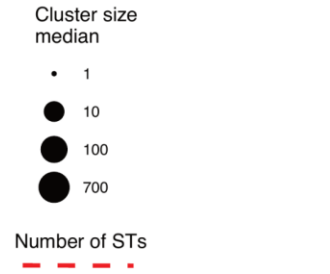
Tool	Similarity	How groups are made
dRep	ANI (Mash first, then ANI)	Mash primary clusters, ANI secondary clusters; cut at 95% (species) or ~99%; best genome per cluster
GeneContRep	Gene content (PanGene-O-Meter)	Greedy: genomes above a gene-content cutoff are grouped, one representative kept
PopPUNK	Core + accessory k-mer distance	Mixture model and network; learns the strain boundary from the data

Same idea, different distance: cut a tree or a network, keep a **representative** (★) per group. The cutoff defines what "the same" means.

Sources: Olm et al. (2017) ISME J 11:2864 (dRep); Ashkenazy and Weigel (2025) (GeneContRep); Lees et al. (2019) Genome Res 29:304 (PopPUNK)

GeneContRep: dereplication by gene content





E. coli, 700 genomes

Same pattern in *P. aeruginosa*, *S. aureus* and *H. pylori*

GeneContRep

Greedy dereplication on gene-content similarity: genomes above a GCS_j cutoff are grouped, and one representative is kept per group.

Where the resolution sits

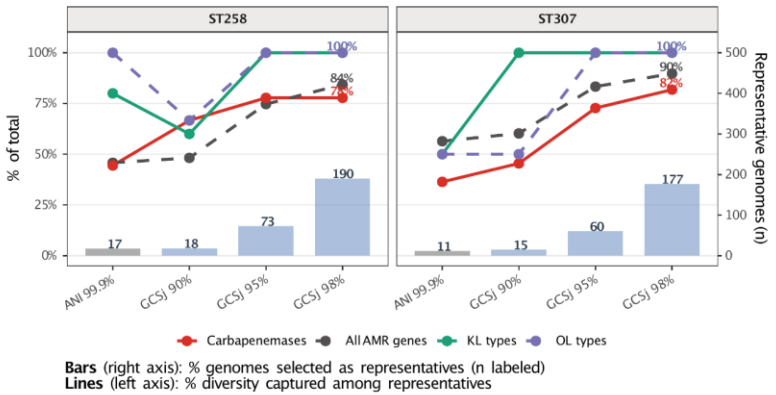
- **Gene content:** spread over a wide, intuitive range of cutoffs
- **ANI:** squeezed into 99–100%
- *Does the extra resolution matter? Next slide.*

Source: Ashkenazy and Weigel (2025), E. coli, 700 genomes

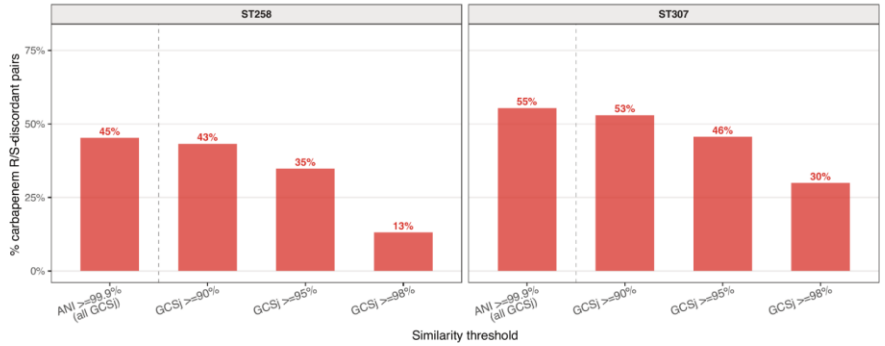
SUPPLEMENTARY · skip if short on time

Gene content keeps the diversity that matters

Representatives kept, and the diversity they capture



Near-identical pairs that disagree on carbapenem resistance



***K. pneumoniae* ST258 & ST307**

~1,000 multidrug-resistant genomes with resistance phenotypes

ANI 99.9% vs. GCS_j 95%

Representatives (ST258 / ST307)	17/11 → 73/60
Capsule types captured (ST307)	100%
Carbapenemase variants (ST307)	82%

Within ANI ≥ 99.9% pairs, a stricter gene-content cutoff cuts carbapenem-discordant pairs from **45%** → **13%** (ST258).

Source: Ashkenazy and Weigel (2025), about 1,000 multidrug-resistant K. pneumoniae genomes

Microbializer: a pangenome tool in the browser

A free **pangenome inference** web server from the Pupko lab (Tel Aviv University): upload genomes, get gene families (the matrix), a tree and annotation. No installation, no login, up to 2,000 genomes.

1 · ORF prediction

Prodigal on each genome (or upload ORFs directly)

2 · Reciprocal best hits

MMseqs2 between genome pairs, with your identity and coverage thresholds

3 · Orthogroups

Bit scores normalised, clustered with MCL; recent paralogs kept in the group

4 · Alignments

MAFFT on proteins, back-translated to codons (RevTrans)

5 · Species tree

Core orthogroups concatenated, IQ-TREE maximum likelihood

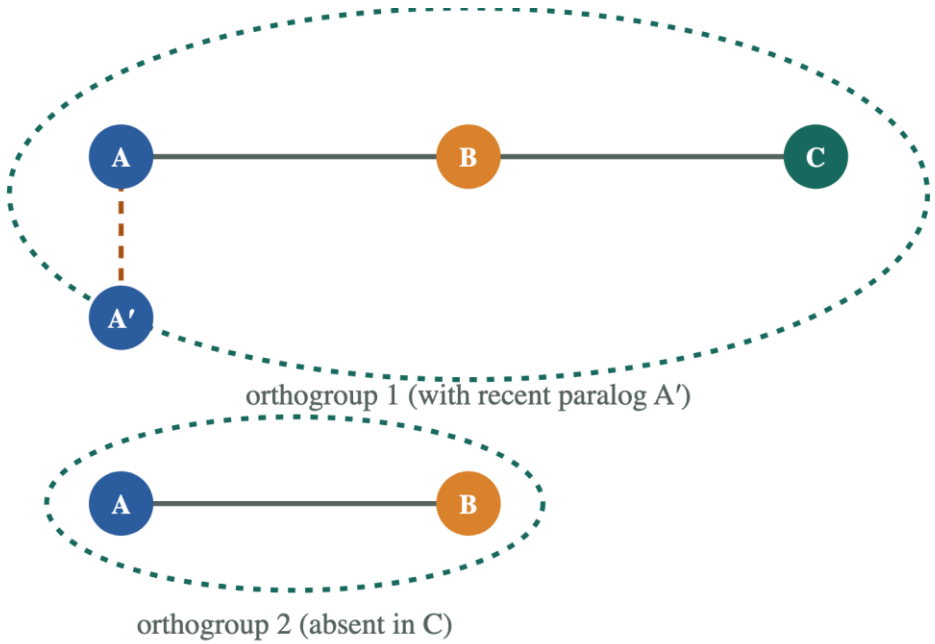
6 · Extras

ANI heatmap, BUSCO-style completeness, KEGG orthology, UMAP of gene content

Link to PanGene-O-Meter: its default builds families with DIAMOND DeepClust (formerly PanX), but it also accepts the Microbializer output.

Source: Shimony et al. (2025) NAR 53:W369; Avram et al. (2019) NAR 47:W88; microbializer.tau.ac.il

How orthogroups are inferred



From hits to orthogroups

- Genes are nodes, **reciprocal best hits** are edges
- Edge weights: normalised bit scores
- **MCL** cuts the graph into dense groups: orthogroups
- Phyletic pattern: 1 if a genome has a member

Parameters that matter: identity and coverage thresholds for a hit; stricter settings give more, smaller families.

Schematic. Method: Shimony et al. (2025) NAR 53:W369 (variant of OrthoMCL, Li et al. 2003)

Everything Microbializer returns

Folder	Content	This workshop
01_ani	Pairwise ANI table and heatmap (FastANI)	Compare with skani
02_orfs	ORFs, translations, GC and ORF-count plots	–
03_genome_completeness	BUSCO-style completeness per genome	Quality control
04_orphan_genes	Genes in no orthogroup	Strain-specific genes
05a_orthogroups	Orthogroup tables (sizes, KO terms), OrthoXML	Annotate modules; PanGene-O-Meter
05b_…_visualizations	Orthogroup sizes, phyletic pattern, matrix plot, UMAP	✓ GLOOME / CoPAP
06_orthogroups_*	Per-orthogroup DNA, protein and alignments	Gene trees, selection
07_aligned_core_*	Core proteome and genome alignments	Basis of the tree
08_species_phylogeny	Maximum-likelihood tree (Newick and image)	✓ GLOOME / CoPAP
09_codon_bias	Codon usage and CAI	–

Source: microbializer.tau.ac.il/overview; one “All outputs” zip contains every folder

The species tree: input for GLOOME and CoPAP

How the tree is made

- Core orthogroups (present in at least the chosen share of genomes)
- Protein alignments concatenated (MAFFT)
- Maximum likelihood with IQ-TREE; more than 1,000 core orthogroups: 1,000 sampled

The tree describes **vertical descent** of the core. Gene content then evolves **on** this tree: that is the GLOOME logic.

Check before GLOOME

- **Root:** choose **outgroup rooting** on the *P. fluorescens* tip (server option; command line: --rooting_method outgroup --leaf NAME). The default is midpoint, which can misplace the root and flip gains and losses
- **Branch lengths:** present in the Newick; GLOOME uses them
- **Names:** tip names must match the headers of the phyletic pattern
- **Format:** Newick starts with “(”, ends with “;”, balanced parentheses; strip support values if rejected

Source: Shimony et al. (2025) NAR 53:W369; microbializer.tau.ac.il; checklist by the author

SUPPLEMENTARY · skip if short on time

Running skani

```
# all-vs-all ANI for a folder of assemblies
conda install -c bioconda skani
skani triangle genomes/*.fna -E -t 8 -o ani_edges.tsv

# one pair
skani dist genomeA.fna genomeB.fna
```

Output (edge list)

- Reference and query names
- **ANI**
- **Aligned fraction** of reference and of query

Reading it

- ≥ 99.9% ANI with high aligned fraction: near-clones
- 95–96% ANI: species boundary
- Low aligned fraction: genomes differ in content, not just sequence

Source: skani documentation (github.com/bluenote-1577/skani); check the column header of your version

SUPPLEMENTARY · skip if short on time

Running PopPUNK

```
# 1. sketch, compute distances
poppunk --create-db --r-files refs.txt \
--output pp_db --threads 8

# 2. fit model, define clusters
poppunk --fit-model bgmm \
--ref-db pp_db --K 4
```

```
# 3. plots, network, tree
poppunk_visualise --ref-db pp_db \
--model-dir pp_db --output pp_vis

# 4. add new genomes later
poppunk_assign --db pp_db \
--query new.txt --output assigned
```

refs.txt: one line per genome, name and path separated by a tab. Always look at the fit plots: the within-strain component should sit near the origin.

Source: PopPUNK documentation (poppunk.readthedocs.io); check option names for your version

Pangenome tools in practice

Tool	Approach	Typical use	Reference
Roary	Clusters annotated genes	One species, close isolates	Page 2015
Panaroo	Fixes annotation errors	One species, draft assemblies	Tonkin-Hill 2020
PanX	Gene clusters + gene trees	Species-level pangenomes	Ding 2018
DIAMOND DeepClust	Fast protein clustering	Thousands of genomes	Buchfink 2023
Microbializer	Reciprocal best hits + MCL	Web server, up to ~2,000	Shimony 2025

PanGene-O-Meter builds its families with DIAMOND DeepClust (default, earlier PanX) and also accepts Microbializer output. Always check identity and coverage thresholds, fragmented genes, paralogs and annotation.

Page et al. 2015 Bioinformatics 31:3691; Tonkin-Hill et al. 2020 Genome Biol 21:180; Ding et al. 2018 NAR 46:e5; Buchfink et al. 2023 bioRxiv, Nat Methods 2026; Shimony et al. 2025 NAR 53:W369

SUPPLEMENTARY · skip if short on time

Running Microbializer

1 · Prepare input

- One FASTA per genome in a **.zip** or .tar.gz
- Draft assemblies are fine; plasmid contigs can be filtered
- Name files by strain: names become tree labels

2 · Set parameters

- Identity and coverage thresholds for orthology
- Core threshold, e.g. 95% of genomes
- Outgroup (we use *P. fluorescens*) to root the tree
- Email for the results link

3 · Submit and wait

- Runs on the Tel Aviv cluster, no account
- Results stay on the server for at least three months
- Keep the result link: it is your run

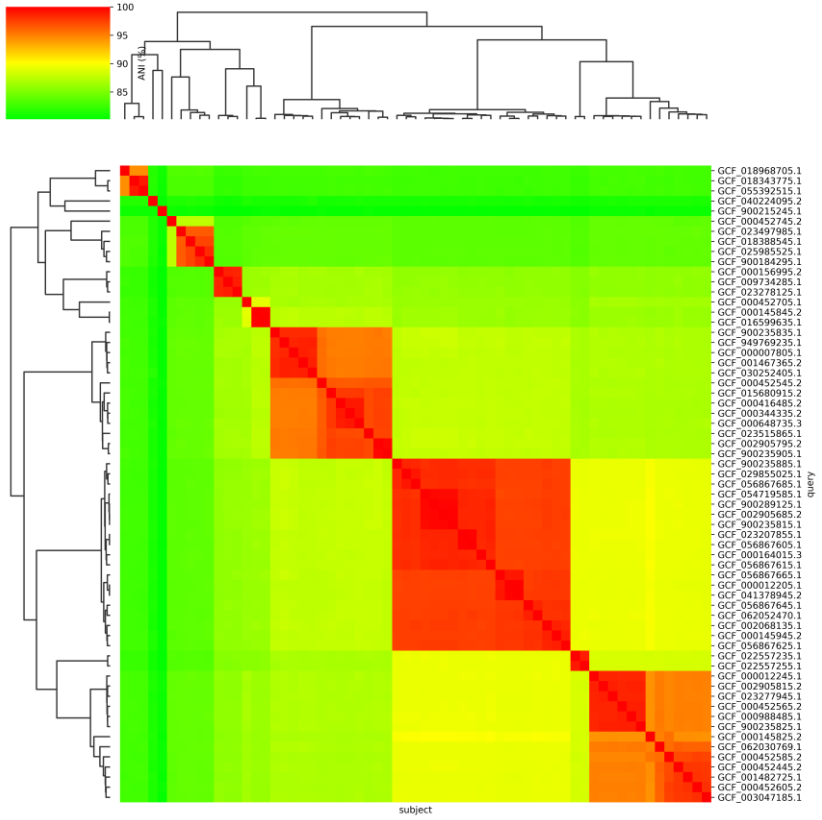
4 · Our Pseudomonas run

63 genomes: *P. syringae* plus a *P. fluorescens* outgroup. We run it once, ahead of time, and reuse it in the afternoon.

microbializer.tau.ac.il

Source: microbializer.tau.ac.il (overview and FAQ); check the exact field labels on the submission form

Our run: 63 Pseudomonas genomes



63
genomes: *P. syringae* + *P. fluorescens* outgroup

26,703
orthogroups

2,362
core (in all 63)

12,536
singletons (in 1 genome)

11,805 families are in 2–62 genomes (shown on the matrix). GLOOME needs no filtering and uses **all 26,703**; CoPAP tests only the variable ones.

ANI heatmap: genomes in blocks. Gene content (next slides) will tell what differs **within** a block.

Real Microbializer output of the workshop set (author's run); ANI = Microbializer 01_ani

Preview of the afternoon hands-on

A · Explore our run

- Open the Pseudomonas result link
- Tree, matrix, orthogroup table
- ANI heatmap and UMAP

B · Gene content

- Gene-content similarity from the orthogroup table
- Dereplicate with GeneContRep
- Compare with ANI

C · Gain, loss, co-evolution

- GLOOME on the phyletic pattern and tree
- Filter, then CoPAP
- Modules with the R script

D · Optional

- GWAS with treeWAS or pyseer
- Or bring your own genomes

Proposed programme; the Pseudomonas run is the shared example

Part 1: take-home messages

Pangenome

A presence/absence matrix of gene families: core, accessory, rare. Inferred, so check thresholds.

Two lenses

ANI (skani) and PopPUNK for who is related; gene content (PanGene-O-Meter) for what they carry.

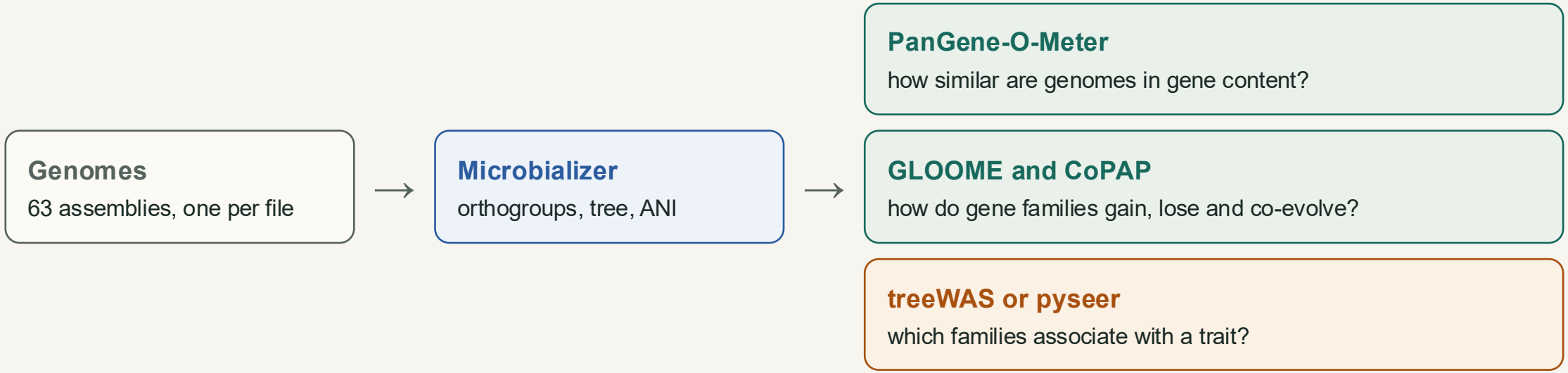
Microbializer

Browser-based pangenome and tree, reused in the hands-on.

The tree matters

Related genomes are not independent: GLOOME, CoPAP, treeWAS and pyseer all account for it.

One matrix, three questions



Every analysis uses the same matrix and the same tree, and every one has to deal with the same problem: **related genomes are not independent**.